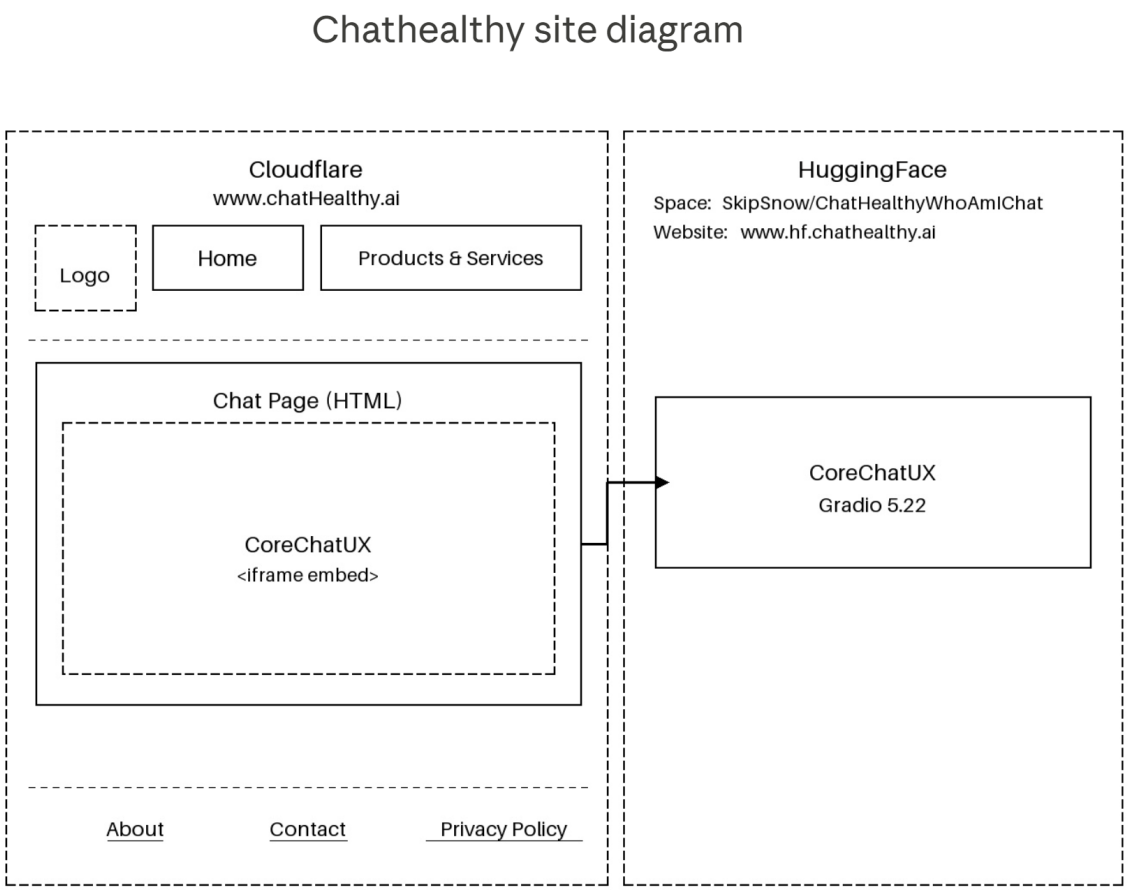
Find Care Architecture

Interface specification used by chat GPT and FIGMA to produce front end and back-end code for the Find Care AI application. Back end uses Gradio and fastAPI.

Overview

- The Find care application demonstrates all the core competencies to solve AI problems using a variety of AI techniques and tools.
- In addition, it is the MVP for an AI application that can help consumers and healthcare professionals to find care for particular types of facilities and problems.
- It also offers hospitals and health insurance companies a starter kit to manage their own networks of providers.

Web site architecture



Frame specification 1

- Overview
  - You received information in the prompt related to this deck about architecture of APIs that must complied with between browser and server.
  - There are two types of fames.
    - Geometric support frames
      - Frames that simply supply geometric support. They have only the functions to change their location and dimension. The location of the frames in terms of their composition of one header with three columns is fixed.
      - The list of geometric support frames from the frame map are:
        - Column 1
        - Column 2 (may be omitted for performance reasons.
        - Column 3
      - Functional frames have responsibilities and must make calls to the server. But can rely on cached data.
      - All of the functional frames are specified in the body of the specification which follows directly.

Frame specification 2

- Header frame:
  - Location at top of app
  - Has a logo. The current one on slide 4 is perfect.
  - Has four links all resolving in a call via the headerFrameAPI to a link resolving service on the server. They are evenly spaced out in the frame page with the logo. They are small and austere in presentation.
    - Secret Sause
      - Mouse over: Tools and processes used to fuel this AI application. (full disclosure)
      - Behavior Go to a secret soft page served by the server.
    - About:
      - Mouse over, about ‘go to the about Find Care page.
      - Behavior: go to a ‘about find care page on the server.

Frame Specification 3

- Contact Find Care
  - Mouse over: ‘Contact Skip Snow (From variable) from Find Care
  - Behavior: Creates a pop up with the default mail client of the user (Title is populated by the server prior to popping up the mail client). If there is no access to such a pop up it pops up the contact email and gives instructions to contact that user. This message is also a variable to be populated by the server. the value coming from the server. Current value is Contact skip Snow, the application’s architect, and implementer, skip.snow@gmail.com.
- Privacy policy
  - Mouse over: Get Find Care’s Privacy Policy
  - Behavior: Go to privacy Policy page.

Frame Specification 4

- Botton manager
  - Button style, raised, and change color when clicking and to another color when waiting on an HTTPS response.
  - Buttons:
    - Insurance portal
      - Mouse over ‘Get context from your insurance portal’
      - Behavior: get’s a window to pop up from the server window asks user to sign off on agreement to get but not retain their identified health data and asks them to fill in the login credentials and URL of their EMR (name of it if it is an app) the submission of the form on this window will be an API of the client and server. Control of the flow must return to the Button manager for all interactions. The server will, if possible log into the portal and get data. In this version only Kaiser’s portal, and UCLA’s portal and Anthem’s portal will be supported, and only if user accounts are set up in these interlaces, and if they support FIHR. Non identified data will be retained.
    - EMR Button:
      - Mouse over: ‘Get history, meds, diagnosis etc. to supply context.
      - Behavior is the same as above but will go to an EMR handler . Only Epic is to be supported in V1.

Frame Specification 4.5

- Session Information
  - Mouse over: ‘gain insight into the advice given and potentially on how to better optimize Find Care’s capabilities.
  - Behavior: A call to the server and a resulting PDF will be returned that the user can download via a link on their browsers. They can also copy formatted text that they could for example paste successfully into Word.
- Choose Model.
  - The user will be able to pick from a variety of LLM models.
    - To do this for some use cases different embedding models must be utilized. Mongo can have many embedded models that the code can choose from at runtime.
    - The list of models and use cases will be presented by the server. Each model will have three data elements:
      - Model name, Description. (model name will be a look up to a user key so much be canonical)
      - Model owner e.g. OpenAI etc.

Frame specification 5

- Session Summary:
  - Function:
    - This control is a display text. Every configurable amount, the server will set as a JS constant, the session context will be sent to a server service that will summarize the session's content on behalf of Find Care. Each time this API is called the component wipes its content an presents the content rendered by the server. The content might have markup.
  - UX:
    - This content must be accessible to the user to copy via system copy modes. Also this control must present a copy icon that will copy the content of the component to the user's clip board. If the user does this, a message of the content must be sent to the server, so this activity can be logged.
    - The control is not scalable. If all the text can not be reasonably rendered, the component must send an error to the server, and display an error message on the screen.

Frame specification 6

- Scrollable output.
  - Ideally this is the Gradio Native Scrollable output component. It must behave as such. It must have a scroll bar, and a copy icon associated with it.
  - Nice to have:
    - Each output will be color coded. With question then indented answer of different font colors. Questions (User prompts) are Green, Answers, what comes back from the Gradio server, Blue.
    - Each Q & A can be copied individually.

Frame specification 7

- Graphic Content:
  - The server can produce interactive content. For example clickable maps where different parts of the map suggest a next action. The server can also serve charts, graphs etc. The images will use the user's questions to figure out a good graphical way to present an answer.
  - Maps
    - A package will be chosen to do map rendering suggest one If you want in the interface catalogue.
  - This might be via a scrollable table as well.
    - In tables, the JS must allow the user normal cell capabilities. For now just editing allowed cells in the table.
  - Bar charts
    - Drillable bar charts might be rendered by the server. This means graphic images, with associated interactive points for drilling
      - The home screen with 50 states where the user can choose a state to start to explore options is an example of this, and the one that the first screen must demonstrate.

Frame Specification 8

- This should be a standard Gradio Prompt interfaces. Files can be uploaded and users can type prompts or past prompts in. If a user paists a prompt to this box, the formatting of the user’s prompt in the clipboard shall be respected. It shall be scrollable and editable for the user. The user’s typing shall not be sent to the server until the user hits a send key.